Date: November 7, 2026

To:
Editor-in-Chief
BMC Medical Imaging
BioMed Central, Springer Nature
London, United Kingdom

Subject: Submission of Original Research Article: *"Topologically Exact 3D/2D Line and Ray Clipping for Distributed CT and MRI Volume Computing: The Analytical Phase-Compensated Bresenham Algorithm"*

Dear Editor-in-Chief,

I am pleased to submit our original research manuscript entitled **"Topologically Exact 3D/2D Line and Ray Clipping for Distributed CT and MRI Volume Computing: The Analytical Phase-Compensated Bresenham Algorithm"** for consideration for publication as a **Research Article** in *BMC Medical Imaging*.

Clinical and Computational Significance for *BMC Medical Imaging*

In modern high-resolution 3D Computed Tomography (CT) and Magnetic Resonance Imaging (MRI), volumetric reconstruction and forward projection require processing massive isotropic grids (512^3 to 1024^3 voxels). To fit within GPU memory and L2 caches, modern clinical reconstruction frameworks partition volumes across distributed GPU nodes, MPI ranks, or tiled cache blocks.

Prevailing volumetric imaging pipelines decouple continuous ray clipping from discrete voxel stepping: continuous trajectories are clipped against sub-volume bounding boxes, and the resulting boundary intersections are rounded to grid coordinates to reseed standard Bresenham steppers. In this paper, we demonstrate that this decoupled approach resets fractional error accumulators to zero, creating a systematic phase shift that disrupts discrete ray stepping. This defect corrupts over 99.7% of boundary-crossing rays, causing boundary seam gaps and missing voxels that generate line-integral attenuation errors up to $15.94\,\mathrm{cm}^{-1}$ (RMSE = 8.9049, PSNR = 14.99 dB on 512 sub-volumes). In full clinical sinogram projections of patient CT and MRI scans, 90.17% to 92.20% of all detector bins are corrupted by phase-shift artifacts, impairing the convergence of iterative algebraic reconstruction techniques (SART, SIRT, MLEM).

Key Contributions

To eliminate these artifacts without communication overhead, this manuscript introduces the **Analytical Phase-Compensated Bresenham (AP-Bresenham)** algorithm:

1. **Closed-Form Diophantine Dynamics:** We derive the exact analytical equations governing multi-dimensional digital straight lines, establishing closed-form formulations for discrete

coordinates and internal accumulator states.

2. **Pure-Integer $O(1)$ Boundary Inversion:** We formulate exact $O(1)$ boundary inversion formulas directly in $\mathbb{Z}$, completely avoiding floating-point rounding errors and iterative searches.

3. **Analytical Historical Phase Injection:** We analytically restore the exact fractional accumulator state at any entry boundary in $O(1)$ time, guaranteeing 100% bit-level equivalence to monolithic traversal without requiring inter-node communication or ghost-cell buffers.

4. **Clinical Patient Validation:** Evaluated across synthetic 3D Shepp-Logan phantoms, multi-compartment brain MRI models, and real clinical acquisitions (human abdominal CT and MNI-152 brain MRI) partitioned into up to 512 sub-volumes, AP-Bresenham achieves a 0.00% seam defect rate, zero corrupted voxels, exact Radon line-integral conservation (RMSE $= 0.0000, \text{PSNR} = \infty$), and a 1.2×–1.6× setup speedup over floating-point clippers.

Declarations and Submission Confirmations

- **Originality:** This manuscript represents original work that has not been previously published and is not under consideration for publication elsewhere.

- **Ethics and Consent:** Not applicable; this study evaluates mathematical phantoms and publicly available, anonymized open-access imaging datasets.

- **Open Source and Reproducibility:** All source code (C++17 and Python), test suites, benchmarks, and clinical projection scripts are open-source and publicly accessible on GitHub at: `https://github.com/hemanthmpkumar/AP-Bresenham`.

- **Competing Interests and Funding:** The author declares no competing financial or non-financial interests. No external funding was received.

Suggested Independent Reviewers

I suggest the following independent expert reviewers with recognized expertise in discrete tomography, volumetric ray tracing, and medical image reconstruction:

1. **Prof. David Coeurjolly**

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None of the proposed reviewers have had collaborative, mentoring, or co-authoring relationships with the author.

Thank you for your consideration of this manuscript.

Sincerely,

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